

IPC144 – Introduction to C

Programming

Practice Exercises

Week 8 - Pointers

Q1 – (Polling) The Internet and the web are enabling more people to network, join a cause, voice opinions, and so on. In this exercise, you'll write a simple polling program that allows users to rate five social-consciousness issues from 1 (least important) to 10 (most important). Pick five causes that are important to you (e.g., political issues, global environmental issues). Use a one-dimensional array `topic` (of type `char *`) to store the five causes. To summarize the survey responses, use a 5-row, 10-column two-dimensional array `responses` (of type `int`), each row corresponding to an element in the `topics` array. When the program runs, it should ask the user to rate each issue. The program displays a summary of the results, including:

- a. A tabular report with the five topics down the left side and the 10 ratings across the top, listing in each column the number of ratings received for each topic.
- b. To the right of each row, show the average of the ratings for that issue.
- c. Which issue received the highest point total? Display both the issue and the point total.
- d. Which issue received the lowest point total? Display both the issue and the point total.

Q2 – (Carbon Footprint Calculator: Arrays of Function Pointers) Using arrays of function pointers, you can specify a set of functions that are called with the same types of arguments and return the same type of data. Governments and companies worldwide are becoming increasingly concerned with carbon footprints (annual releases of carbon dioxide into the atmosphere) from buildings burning various types of fuels for heat, vehicles burning fuels for power, and the like. Many scientists blame these greenhouse gases for the phenomenon called global warming.

Create three functions that help calculate the carbon footprint of a building, a car and a bicycle, respectively. Each function should input appropriate data from the user, then calculate and display the carbon footprint. (Check out a few websites that explain how to calculate carbon footprints.)

Each function should receive no parameters and return void. Write a program that prompts the user to enter the type of carbon footprint to calculate, then calls the corresponding function in the array of function pointers. For each type of carbon footprint, display some identifying information and the object's carbon footprint.